SECTION A – 1 MARK QUESTIONS (10 × 1 = 10 Marks)

1. What is the primary advantage of LoRA over full fine-tuning?

a) Higher accuracy
b) Parameter efficiency (90% memory reduction)
c) Faster inference
d) Better generalization

2. Which of these is a data preprocessing step in model training?

a) Model architecture design
b) Dataset cleaning and normalization
c) Hyperparameter tuning
d) Loss function selection

3. What does PEFT stand for?

a) Prompt Engineering For Training
b) Parameter-Efficient Fine-Tuning
c) Parallel Efficient Feed-forward Training
d) Post-Evaluation Fine-Tuning

4. Which evaluation metric measures recall of word sequences in summarization?

a) BLEU
b) ROUGE
c) Perplexity
d) Accuracy

5. What is stratified cross-validation used for?

a) Increase training speed
b) Maintain class distribution in train/test splits
c) Reduce overfitting
d) Improve model accuracy

6. How should you handle missing values in training data?

a) Ignore them
b) Delete the entire sample
c) Impute or delete strategically
d) Use random values

7. What is the primary goal of hyperparameter tuning?

a) Train faster
b) Reduce model size
c) Find optimal model configuration
d) Eliminate overfitting

8. Which technique helps prevent overfitting?

a) Increasing model complexity
b) Using more training data
c) Reducing features
d) All of above except a)

9. What is batch size in model training?

a) Number of epochs
b) Total number of samples
c) Number of samples processed before weight update
d) Learning rate value

10. How does early stopping prevent overfitting?

a) Reduces learning rate gradually
b) Stops training when validation loss stops improving
c) Removes difficult samples
d) Increases regularization

SECTION B – 5 MARK QUESTIONS Answer ANY 6 (6 × 5 = 30 Marks)

1. Explain neural networks and their role in Generative AI.

2. Define tokens and embeddings with examples.

3. What is diffusion model? Explain briefly.

4. Explain role-based prompting.

5. What is data cleaning in AI model training?

6. Write short notes on Generative Adversarial Networks (GANs).

7. What is semantic similarity in AI?

SECTION C – 10 MARK QUESTIONS Answer ANY 4 (4 × 10 = 40 Marks)

1. Explain the architecture of Large Language Models.

2. Describe the steps involved in dataset preparation.

3. Explain RAG architecture with diagram explanation.

4. Explain multimodal AI systems with examples.

5. Discuss AI bias and fairness in model training.

SECTION D – 20 MARK CASE STUDY Answer ANY 1 (1 × 20 = 20 Marks)

CASE STUDY 1: E-COMMERCE AI RECOMMENDATION ENGINE

SCENARIO:

TechMart, a major e-commerce platform, deploys a Generative AI recommendation engine to suggest products to customers. The system uses LLMs and embeddings to understand customer preferences and product characteristics. After 3 months:

- Users report irrelevant product recommendations
- Minority customers receive fewer product suggestions
- The system occasionally generates misleading product descriptions
- High computational costs due to inefficient model deployment
- Data privacy concerns regarding customer purchase history storage

QUESTIONS

Q1 Analyze the root causes of irrelevant recommendations and explain how RAG architecture could improve accuracy.
Q2 Identify the fairness issues in the system and propose mitigation strategies to ensure equitable recommendations across demographics.
Q3 Design a data privacy framework ensuring GDPR compliance while maintaining recommendation quality.
Q4 Suggest optimization techniques to reduce computational costs while maintaining model performance using LoRA/PEFT.

CASE STUDY 2: HEALTHCARE AI DIAGNOSTIC ASSISTANT

SCENARIO:

MediCare Hospital implements an AI diagnostic assistant powered by multimodal LLMs to analyze medical images and patient history. The system assists doctors in diagnosis. After deployment:

- AI provides incorrect diagnoses in 8% of cases
- System fails to flag critical conditions in senior patients
- Unauthorized access to patient data through prompt injection attacks
- Lack of explainability in AI decisions affecting doctor trust
- No audit trail for liability and accountability purposes

Q1 Discuss ethical concerns and develop a human-in-the-loop verification system with clear escalation protocols.

Q2 Design comprehensive security measures to prevent prompt injection attacks and ensure data confidentiality with encryption strategies.

Q3 Propose evaluation metrics and bias detection methods to ensure fair diagnosis across age groups and demographics.

Q4 Create a compliance and incident response plan meeting HIPAA, GDPR, and medical liability requirements.